

```
In [2]: import pandas as pd
import matplotlib.pyplot as pl
import seaborn as sns
```

```
In [3]: data = pd.read_csv(r'C:\Users\LORD\OneDrive\Desktop\python
files\u_food_marketing.csv')
data2 = data
```

```
In [4]: pd.set_option('display.max.columns',45)
data.head(2)
```

```
Out[4]:
```

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts
0	58138.0	0	0	58	635	88	546	172
1	46344.0	1	1	38	11	1	6	2

```
In [5]: data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2205 entries, 0 to 2204
Data columns (total 39 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Income                                2205 non-null   float64
1   Kidhome                               2205 non-null   int64
2   Teenhome                              2205 non-null   int64
3   Recency                               2205 non-null   int64
4   MntWines                              2205 non-null   int64
5   MntFruits                             2205 non-null   int64
6   MntMeatProducts                       2205 non-null   int64
7   MntFishProducts                       2205 non-null   int64
8   MntSweetProducts                      2205 non-null   int64
9   MntGoldProds                          2205 non-null   int64
10  NumDealsPurchases                     2205 non-null   int64
11  NumWebPurchases                       2205 non-null   int64
12  NumCatalogPurchases                   2205 non-null   int64
13  NumStorePurchases                     2205 non-null   int64
14  NumWebVisitsMonth                     2205 non-null   int64
15  AcceptedCmp3                          2205 non-null   int64
16  AcceptedCmp4                          2205 non-null   int64
17  AcceptedCmp5                          2205 non-null   int64
18  AcceptedCmp1                          2205 non-null   int64
19  AcceptedCmp2                          2205 non-null   int64
20  Complain                              2205 non-null   int64
21  Z_CostContact                         2205 non-null   int64
22  Z_Revenue                             2205 non-null   int64
23  Response                              2205 non-null   int64
24  Age                                    2205 non-null   int64
25  Customer_Days                         2205 non-null   int64
26  marital_Divorced                      2205 non-null   int64
27  marital_Married                       2205 non-null   int64
28  marital_Single                        2205 non-null   int64
29  marital_Together                      2205 non-null   int64
30  marital_Widow                         2205 non-null   int64
31  education_2n Cycle                    2205 non-null   int64
32  education_Basic                       2205 non-null   int64
33  education_Graduation                  2205 non-null   int64
34  education_Master                      2205 non-null   int64
35  education_PhD                         2205 non-null   int64
36  MntTotal                              2205 non-null   int64
37  MntRegularProds                      2205 non-null   int64
38  AcceptedCmpOverall                    2205 non-null   int64
dtypes: float64(1), int64(38)
memory usage: 672.0 KB

```

```
In [6]: data.drop_duplicates(keep = False ,inplace = True)
```

```
In [7]: data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 1843 entries, 0 to 2204
Data columns (total 39 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   Income                               1843 non-null   float64
 1   Kidhome                             1843 non-null   int64  
 2   Teenhome                            1843 non-null   int64  
 3   Recency                             1843 non-null   int64  
 4   MntWines                            1843 non-null   int64  
 5   MntFruits                           1843 non-null   int64  
 6   MntMeatProducts                     1843 non-null   int64  
 7   MntFishProducts                     1843 non-null   int64  
 8   MntSweetProducts                    1843 non-null   int64  
 9   MntGoldProds                       1843 non-null   int64  
10   NumDealsPurchases                   1843 non-null   int64  
11   NumWebPurchases                     1843 non-null   int64  
12   NumCatalogPurchases                 1843 non-null   int64  
13   NumStorePurchases                   1843 non-null   int64  
14   NumWebVisitsMonth                   1843 non-null   int64  
15   AcceptedCmp3                        1843 non-null   int64  
16   AcceptedCmp4                        1843 non-null   int64  
17   AcceptedCmp5                        1843 non-null   int64  
18   AcceptedCmp1                        1843 non-null   int64  
19   AcceptedCmp2                        1843 non-null   int64  
20   Complain                            1843 non-null   int64  
21   Z_CostContact                       1843 non-null   int64  
22   Z_Revenue                           1843 non-null   int64  
23   Response                            1843 non-null   int64  
24   Age                                 1843 non-null   int64  
25   Customer_Days                       1843 non-null   int64  
26   marital_Divorced                    1843 non-null   int64  
27   marital_Married                     1843 non-null   int64  
28   marital_Single                      1843 non-null   int64  
29   marital_Together                    1843 non-null   int64  
30   marital_Widow                       1843 non-null   int64  
31   education_2n Cycle                  1843 non-null   int64  
32   education_Basic                     1843 non-null   int64  
33   education_Graduation                1843 non-null   int64  
34   education_Master                    1843 non-null   int64  
35   education_PhD                       1843 non-null   int64  
36   MntTotal                            1843 non-null   int64  
37   MntRegularProds                     1843 non-null   int64  
38   AcceptedCmpOverall                  1843 non-null   int64  
dtypes: float64(1), int64(38)
memory usage: 575.9 KB

```

```
In [8]: data['Kids_home'] = data[['Kidhome', 'Teenhome']].sum(axis = 1)
```

Reference Table marital_states

```
In [9]: data['marital_Divorced'] = data['marital_Divorced'].replace({1:5,0:0})
data['marital_Married'] = data['marital_Married'].replace({1:4,0:0})
data['marital_Single'] = data['marital_Single'].replace({1:3,0:0})
data['marital_Together'] = data['marital_Together'].replace({1:2,0:0})
data['marital_Widow'] = data['marital_Widow'].replace({1:1,0:0})
```

```
In [10]: data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 1843 entries, 0 to 2204
Data columns (total 40 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Income                                1843 non-null   float64
 1   Kidhome                               1843 non-null   int64
 2   Teenhome                              1843 non-null   int64
 3   Recency                               1843 non-null   int64
 4   MntWines                              1843 non-null   int64
 5   MntFruits                             1843 non-null   int64
 6   MntMeatProducts                       1843 non-null   int64
 7   MntFishProducts                       1843 non-null   int64
 8   MntSweetProducts                      1843 non-null   int64
 9   MntGoldProds                          1843 non-null   int64
10   NumDealsPurchases                     1843 non-null   int64
11   NumWebPurchases                       1843 non-null   int64
12   NumCatalogPurchases                  1843 non-null   int64
13   NumStorePurchases                    1843 non-null   int64
14   NumWebVisitsMonth                     1843 non-null   int64
15   AcceptedCmp3                          1843 non-null   int64
16   AcceptedCmp4                          1843 non-null   int64
17   AcceptedCmp5                          1843 non-null   int64
18   AcceptedCmp1                          1843 non-null   int64
19   AcceptedCmp2                          1843 non-null   int64
20   Complain                              1843 non-null   int64
21   Z_CostContact                         1843 non-null   int64
22   Z_Revenue                             1843 non-null   int64
23   Response                              1843 non-null   int64
24   Age                                    1843 non-null   int64
25   Customer_Days                         1843 non-null   int64
26   marital_Divorced                      1843 non-null   int64
27   marital_Married                       1843 non-null   int64
28   marital_Single                        1843 non-null   int64
29   marital_Together                      1843 non-null   int64
30   marital_Widow                         1843 non-null   int64
31   education_2n Cycle                    1843 non-null   int64
32   education_Basic                       1843 non-null   int64
33   education_Graduation                  1843 non-null   int64
34   education_Master                      1843 non-null   int64
35   education_PhD                         1843 non-null   int64
36   MntTotal                              1843 non-null   int64
37   MntRegularProds                       1843 non-null   int64
38   AcceptedCmpOverall                    1843 non-null   int64
39   Kids_home                             1843 non-null   int64
dtypes: float64(1), int64(39)
memory usage: 590.3 KB

```

```

In [11]: data['marital_states']
         =data[['marital_Divorced','marital_Married','marital_Single','marital_Together',
         = 1)

```

```
In [12]: data.drop(['marital_Divorced','marital_Married','marital_Single','marital_Together'],
                  =1)
```

```
Out[12]:
```

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts
0	58138.0	0	0	58	635	88	546	
1	46344.0	1	1	38	11	1	6	
2	71613.0	0	0	26	426	49	127	
3	26646.0	1	0	26	11	4	20	
4	58293.0	1	0	94	173	43	118	
...	...	...	...	...	...	...	...	
2198	26816.0	0	0	50	5	1	6	
2200	61223.0	0	1	46	709	43	182	
2202	56981.0	0	0	91	908	48	217	
2203	69245.0	0	1	8	428	30	214	
2204	52869.0	1	1	40	84	3	61	

1843 rows × 36 columns

```
In [13]: data['marital_states_str'] =
data['marital_states'].map({1:'Widow',2:'Together',3:'Single',4:'Married',5:'Divorced'})
```

```
In [14]: data['education_2n Cycle'] = data['education_2n Cycle'].replace({1:1,0:0})
data['education_Basic'] = data['education_Basic'].replace({1:2,0:0})
data['education_Graduation'] =
data['education_Graduation'].replace({1:3,0:0})
data['education_Master'] = data['education_Master'].replace({1:4,0:0})
data['education_PhD'] = data['education_PhD'].replace({1:5,0:0})
```

```
In [15]: data.head()
```

Out[15]:

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts
0	58138.0	0	0	58	635	88	546	172
1	46344.0	1	1	38	11	1	6	2
2	71613.0	0	0	26	426	49	127	111
3	26646.0	1	0	26	11	4	20	10
4	58293.0	1	0	94	173	43	118	46

In [18]:

```
data['Accepted_cmp'] = data[['AcceptedCmp1',  
                             'AcceptedCmp2', 'AcceptedCmp5', 'AcceptedCmp4', 'AcceptedCmp3', 'Response']].ast  
                             = 1)
```

In [19]:

```
data[data['Accepted_cmp'] != 0].head()
```

Out[19]:

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts
0	58138.0	0	0	58	635	88	546	172
8	30351.0	1	0	19	14	0	24	3
9	5648.0	1	1	68	28	0	6	7
17	76995.0	0	1	91	1012	80	498	0
21	58607.0	0	1	63	867	0	86	0

In [20]:

```
age_asc = data['Age'].sort_values(ascending = False)  
age_asc
```

Out[20]:

```
1924    80  
407     79  
1130    77  
1897    77  
398     77  
      ..  
727     25  
1824    25  
2180    25  
1150    24  
43      24  
Name: Age, Length: 1843, dtype: int64
```

In [21]:

```
age_groups = [(20,30),(31,40),(41,50),(51,60),(61,70),(71,85)]  
  
def age_grouping(Age):
```

```

for i in age_groups:
    if i[0] <= Age <= i[1] :
        return f"{i[0]}-{i[1]}"
    return "UNKNOWN"

data['Age_group'] = data['Age'].apply(age_grouping)

```

In [22]: `data.head(4)`

Out[22]:

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts
0	58138.0	0	0	58	635	88	546	172
1	46344.0	1	1	38	11	1	6	2
2	71613.0	0	0	26	426	49	127	111
3	26646.0	1	0	26	11	4	20	10

In [50]:

```

counts = data['Age_group'].value_counts()
percentage = counts / data.shape[0]
percentage_data = percentage.reset_index()

```

In [56]:

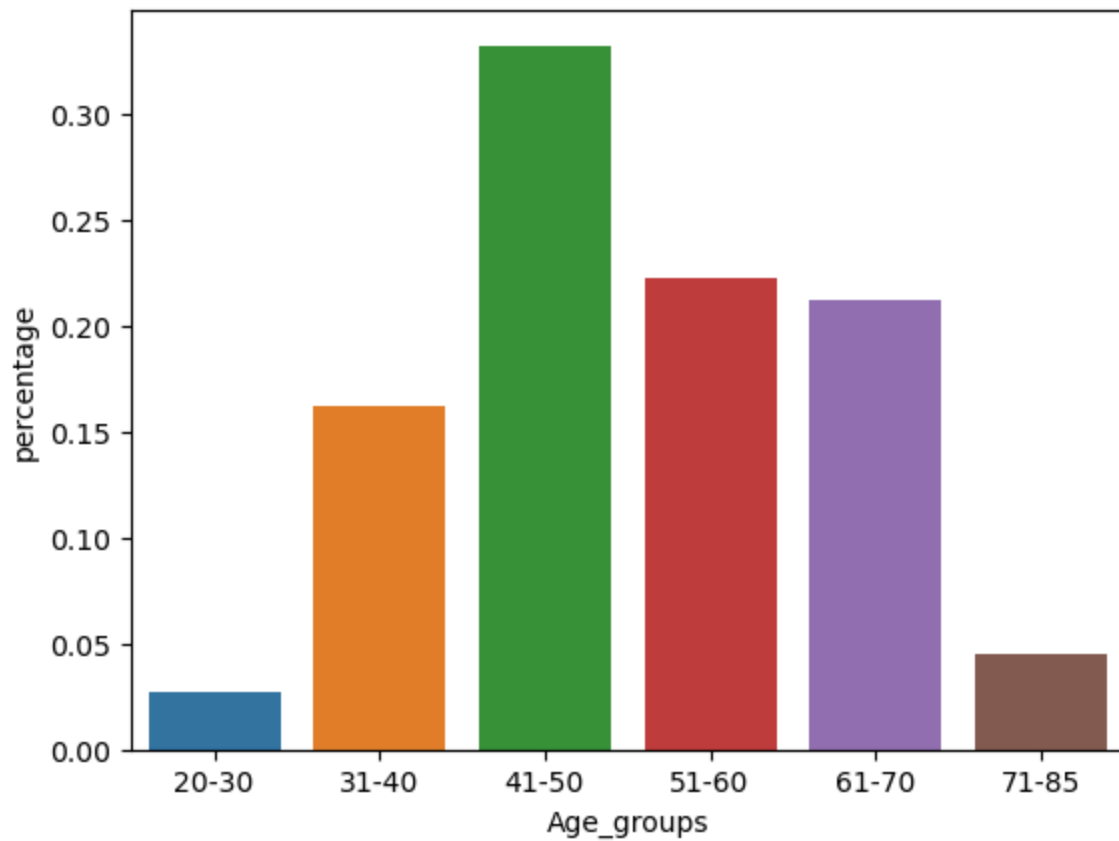
```

percentage_data.columns=['Age_groups','percentage']
percentage_vizu = percentage_data.sort_values('Age_groups')

```

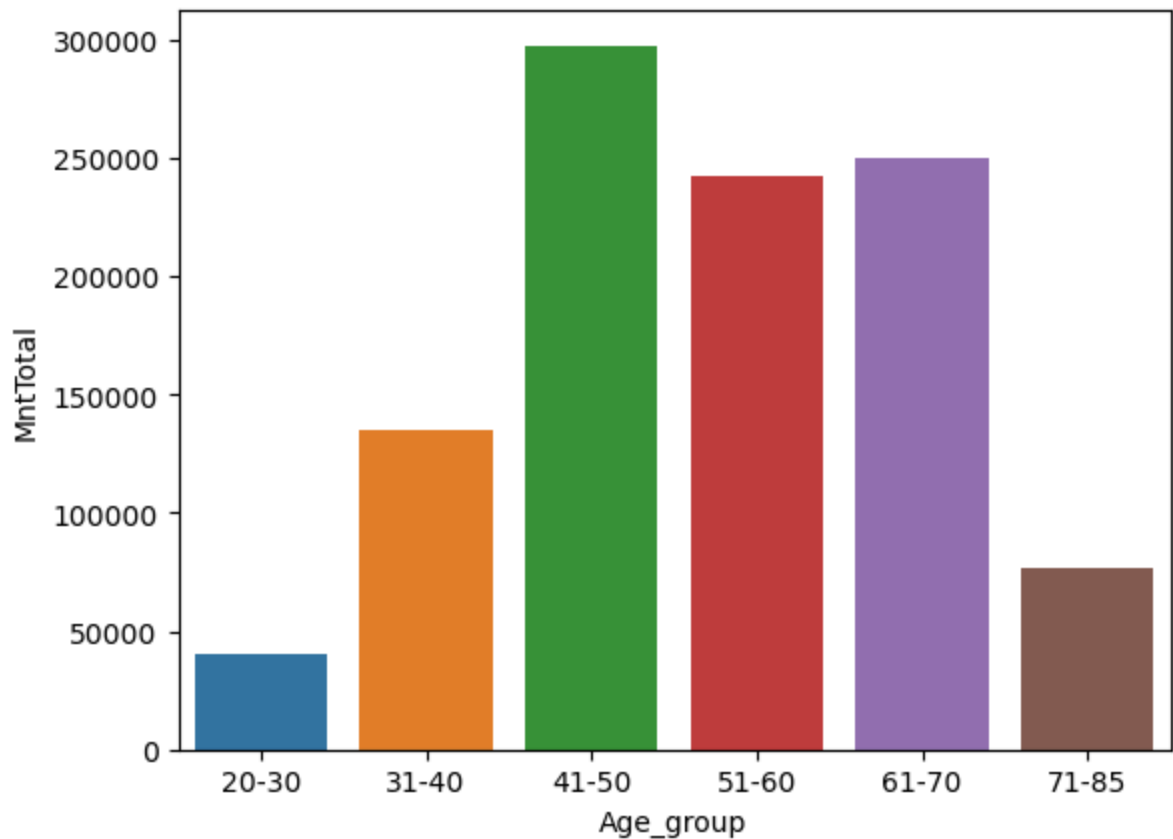
In [58]: `sns.barplot(data = percentage_vizu , x = 'Age_groups', y = 'percentage' )`

Out[58]: `<Axes: xlabel='Age_groups', ylabel='percentage'>`



```
In [64]: total_spent = data.groupby('Age_group')['MntTotal'].sum().reset_index()
sns.barplot(data = total_spent , x = 'Age_group' , y = 'MntTotal')
```

```
Out[64]: <Axes: xlabel='Age_group', ylabel='MntTotal'>
```



In [60]: `data.head(2)`

Out[60]:

	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts
0	58138.0	0	0	58	635	88	546	172
1	46344.0	1	1	38	11	1	6	2

In [101]:

```

type_of_purchases =
pd.DataFrame(data[['NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth'],
                  columns = ['Sums']])
total_types = type_of_purchases.reset_index()
total_types

```

Out[101]:

	index	Sums
0	NumCatalogPurchases	4871
1	NumStorePurchases	10675
2	NumWebVisitsMonth	9853

In [89]: `total_types = total_types.rename(columns = {'index' : 'Types_of_purchases'})`

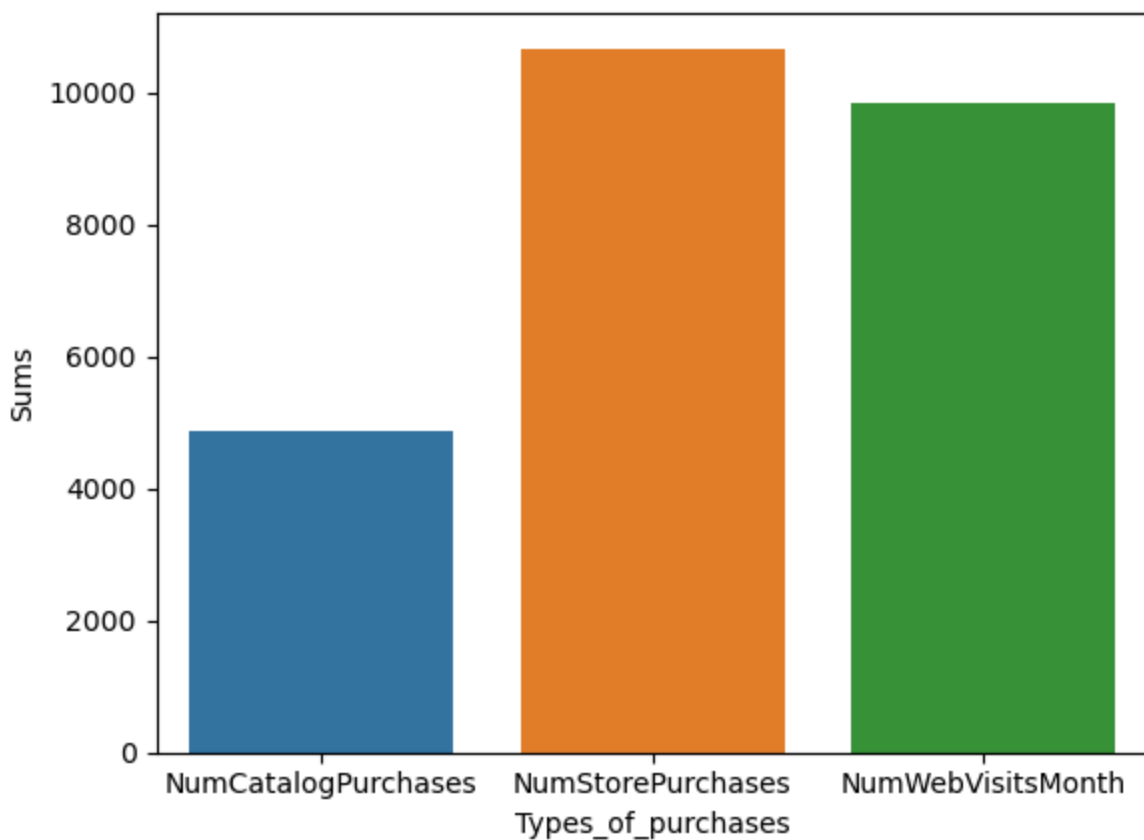
```
In [90]: total_types
```

```
Out[90]:
```

	Types_of_purchases	Sums
0	NumCatalogPurchases	4871
1	NumStorePurchases	10675
2	NumWebVisitsMonth	9853

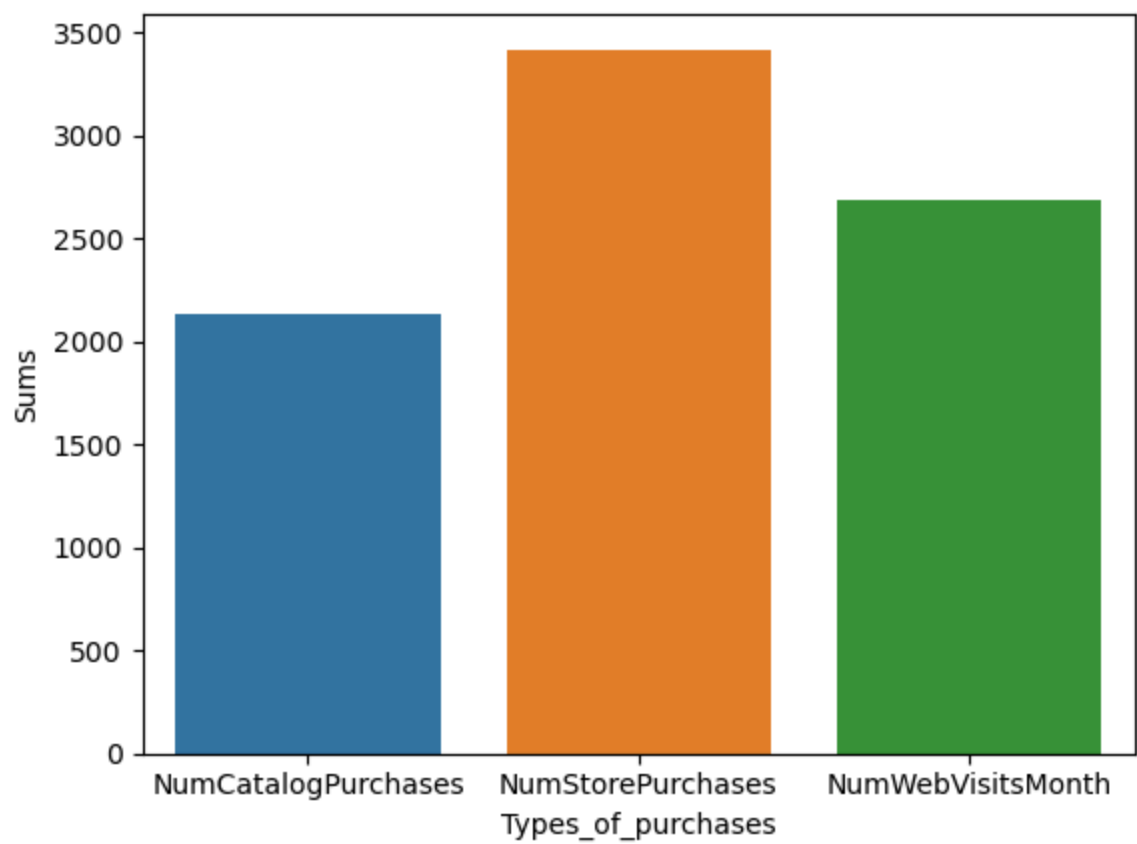
```
In [91]: sns.barplot(data = total_types , x = 'Types_of_purchases' , y = 'Sums')
```

```
Out[91]: <Axes: xlabel='Types_of_purchases', ylabel='Sums'>
```



```
In [110... accept_cmp = data[data['Accepted_cmp'] != 0]
type_of_purchases2 =
pd.DataFrame(accept_cmp[['NumCatalogPurchases', 'NumStorePurchases', 'NumWebVi
= ['Sums']])
type_of_purchases2
total_types2 = type_of_purchases2.reset_index()
total_types2 = total_types2.rename(columns = {'index' :
'Types_of_purchases'})
sns.barplot(data = total_types2 , x = 'Types_of_purchases' , y = 'Sums')
```

Out[110]: <Axes: xlabel='Types_of_purchases', ylabel='Sums'>



In []:

In []:

In []:

In []:

In []:

In []:

In []: